



Optimizing Credit Card Approvals at SecureBank Through Risk Segmentation



Profile

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FSDA October '25
Batch Amsterdam - Team 6



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**“The essence of
investment management
is the management of
risks, not the
management of returns.”**

—Benjamin Graham

Executive Summary

PROBLEM STATEMENT

Despite a **low overall default rate (0.87%)**, some applicants falls into a high-risk category about **(10,9%)**.



KEY FINDINGS

- **Income is flat:** default rates are similar across bands.
- Stability is key: **unemployed applicants have 33% higher risk.**
- Burden matters: **large families (5+ members) spike in default (>1%).**



OBJECTIVE

Shift from **an income-only strategy to a multi-variable segmentation analysis** to **reduce high-risk approvals by 10% within the next quarter.**



RECOMMENDATION

- **Stop tightening income rules;** it rejects good customers.
- Penalize the **“Toxic Trio”:** **Unemployed + Low Income + Large Family.**
- **Implement segment-based risk multipliers** in approvals.



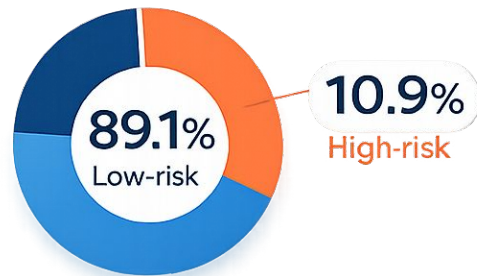
Project Background

About SecureBank

SecureBank is a financial institution that provides **credit card products** to retail customers. As part of its growth strategy, SecureBank aims to expand its customer base while maintaining a healthy credit portfolio.

- However, recent internal portfolio reviews show that although **the overall default rate remains low (0.8%)**, risk is highly concentrated in specific customer segments.
- Approximately **10.9% of approved applicants** fall into high-risk categories, indicating that a small group of customers contributes disproportionately to potential losses.

Risk Concentration



1 in 9 approved applicants
is high-risk



Project Goals



Business Problem

Within the **next quarter**, this project aims to improve credit card approval decisions by identifying key applicant characteristics that differentiate default and non-default customers based on historical credit records, with the objective of **reducing high-risk applicant approvals by at least 10%** through data-driven and explainable credit scoring analysis.



Objective

This project aims to **improve credit risk screening by identifying key applicant attributes** that differentiate good and bad clients, with the objective of **reducing high-risk applicant approvals by at least 10%** through data-driven and explainable credit scoring analysis.

Data Overview and Project of Scope

Data Source



This [dataset](#) consist **438,557** applicants

Merged from:

- **Application records**
- **Historical credit records**

Contains **demographic, financial, and behavioral attributes** for retail credit card customers.

Key Variables

The analysis focuses on **three** main variable groups:

- **Applicant Demographics**
- **Applicant Financial Stability**
- **Applicant Credit Behavior**

Project Scope

This project aims to:

- **Identify characteristics** that differentiate **good vs bad customers**
- **Measure risk concentration** across segments
- **Support data-driven approval optimization**

Research Methodologies

Business Understanding

Define objectives and business problem.



Align stakeholders on key performance indicators (KPIs)

Data Preparation & Cleaning

Merge, clean and preprocess data.



Ensure data integrity for downstream analysis

Exploratory Data Analysis

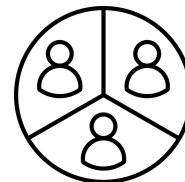
Conduct descriptive and statistical analysis.



Uncover hidden patterns and correlations

Risk-Based Segmentation Analysis

Identify customer segments by risk level.



Profile customers based on predictive risk scores

Insight & Recommendation

Provide data-driven strategies to minimize risk.



Actionable roadmaps for business mitigation

Exploratory Data Analysis (EDA) & Findings

Applicant Profile & Statistical Findings



160,780

MEDIAN ANNUAL INCOME



43 Years

MEDIAN APPLICANT AGE



5.3 Years

MEDIAN EMPLOYMENT LENGTH

Metric	AMT_INCOME_TOTAL	AGE_YEARS	EMPLOYED_YEARS	CNT_CHILDREN	CNT_FAM_MEMBERS
Count	438,557	438,557	363,228	438,557	438,557
Mean	187,524.29	43.83	7.19	0.43	2.19
Std	110,086.85	11.47	6.57	0.72	0.90
Min	26,100	21	0	0	1
25%	121,500	34	2.5	0	2
50%	160,780	43	5.3	0	2
75%	225,000	53	9.6	1	3
Max	6,750,000	69	48	19	20

- Applicant population has a median annual income of **160,780**, showing a right-skewed distribution.
- Median age is **43 years**, reflecting a stable, working-age demographic.
- Most applicants possess relatively **short employment histories (Median 5.3 years)**.
- Low variation in child count; majority live in **small households** (2-3 family members).
- Extreme values in income suggest **high-net-worth outliers** that require specialized risk handling.

Applicants Occupation

Occupation Type	Count
Unknown	134,203
Laborers	78,240
Core staff	43,007
Sales staff	41,098
Managers	35,487
Drivers	26,090
High skill tech staff	17,289
Accountants	15,985
Medicine staff	13,520
Cooking staff	8,076

- A significant portion of applicants fall under the **“Unknown”** category, reflecting missing job information.
- Among known occupations, **laborers, sales staff, and core staff** are the most common.
- **Managerial and high-skill positions represent a smaller** share of the applicant pool.
- The distribution highlights **diverse employment profiles** with varying income levels.

Applicant Education

Education Level	Percentage
Secondary / secondary special	68.82%
Higher education	26.80%
Incomplete higher	3.39%
Lower secondary	0.92%
Academic degree	0.07%

- The majority of applicants have **secondary or secondary special education (68%)**.
- Around **27% of applicants hold higher education degrees**.
- Advanced academic degrees represent only **a very small fraction** of the applicant population.
- This indicates that the applicant base is dominated by **mid-level educational backgrounds**.

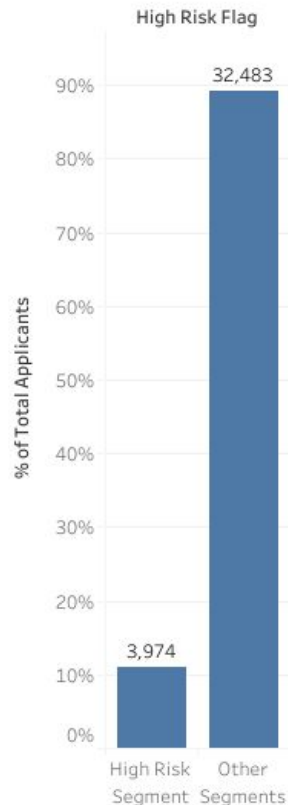
Credit Status EDA

Status	Count
C	442,031
0	383,120
X	209,230
1	11,090
5	1,693
2	868
3	320
4	223

- The majority of credit records fall under **“C” (paid off)** and **“0” (1-29 days past due)** statuses.
- Records indicating **no active loan (“X”)** also represent a substantial portion of the data.
- Severe delinquency statuses (**3, 4, and 5**) appear **very infrequently**, indicating that extreme payment issues are relatively rare at the record level.

Credit Status EDA

Policy Impact Summary

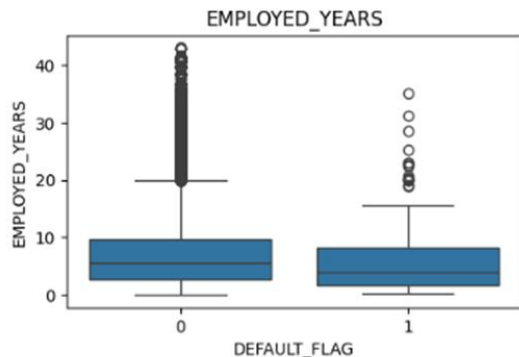


- The bar chart visually confirms that **non-default** and **low-delinquency** statuses **dominate** the dataset.
- **High-risk delinquency statuses** form a very small tail, emphasizing the **need for careful aggregation at the customer level** rather than **relying on individual monthly records**.

Advanced Exploratory Data Analysis (EDA) in Applicant Segmentation

“Default risk is primarily driven by employment stability”

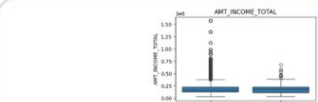
Key Differentiator
EMPLOYED_YEARS



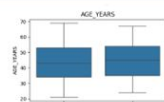
- Employed years shows the **clearest separation between non-default and default customers.**
- **Income** and **age** exhibit **substantial overlap**, indicating **weak discriminatory power** among already approved applicants.
- Household size provides **minimal differentiation.**

Implication

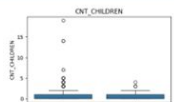
If approval decisions rely too **heavily on income**, applicants with **sufficient earnings but unstable employment histories** may still be approved.



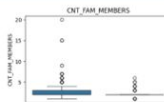
AMT_INCOME_TOTAL



AGE_YEARS



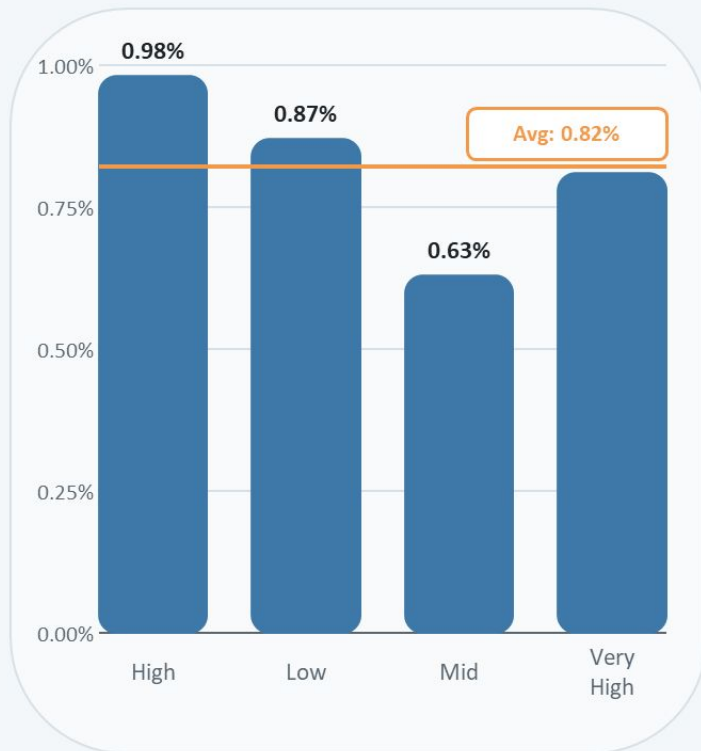
CNT_CHILDREN



CNT_FAM_MEMBERS

“Income Alone Does Not Differentiate Credit Risk”

Default Rate by Income Segments



Approved applicants only • Default rate (%)

- Default rates are **relatively similar across income segments**.
- High-income applicants show **slightly higher default than middle-income groups** — income alone is not a **reliable risk discriminator** within approved users.

Implication: prioritize employment stability + repayment behavior features over income-only rules.

“Unemployed Applicants Are Substantially More Likely to Default”

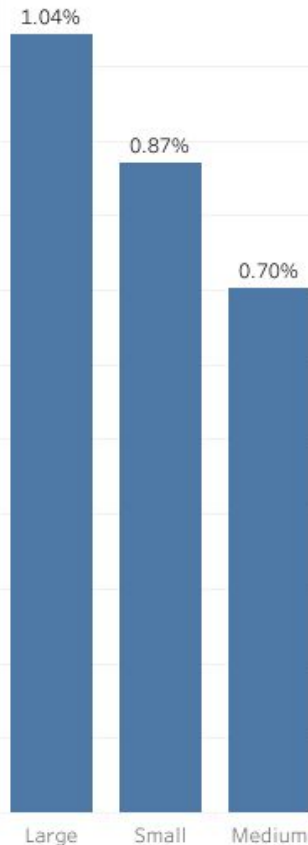
Default Rate by
Employment
Status



- **Unemployed applicants** have **higher default rate** compared to employed applicants.
- **Employment stability** is a **meaningful driver of default risk**, even among customers who were already approved.

Implication: Tighten underwriting for unemployed applicants and prioritize employment stability as a key screening signal to reduce defaults.

Default Rate by
Family Segments



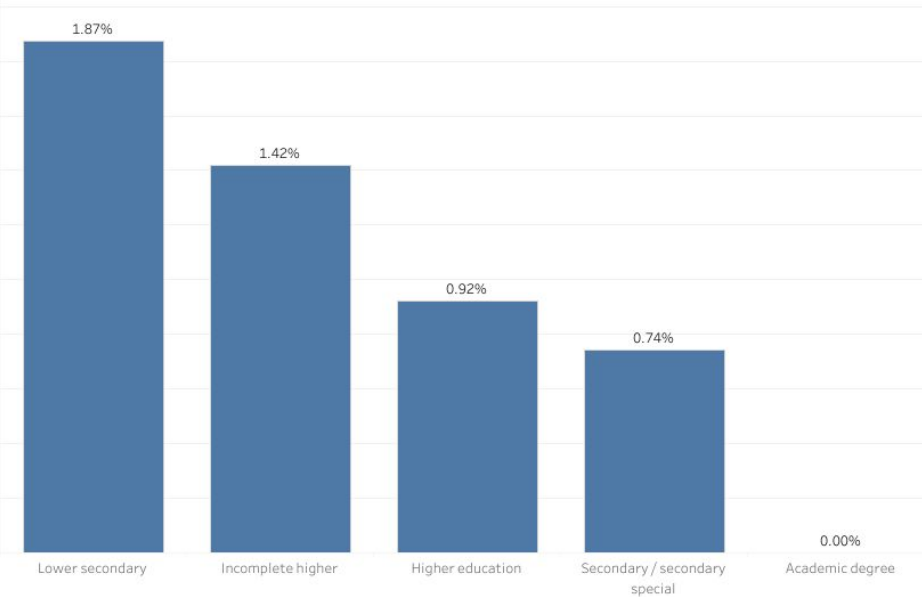
“Larger Households Face Higher Default Risk”

- Default risk is lowest for **small-to-medium households (2–4 members)** and rises for **larger families (5–6 members)**, suggesting that household burden **increases financial stress** and **default risk**.

Implication: Household size should be treated as an affordability risk signal then larger households may have tighter disposable income and higher repayment stress, so underwriting should apply additional affordability checks

“Default Risk Declines as Education Level Increases”

Default Rate by Education Levels

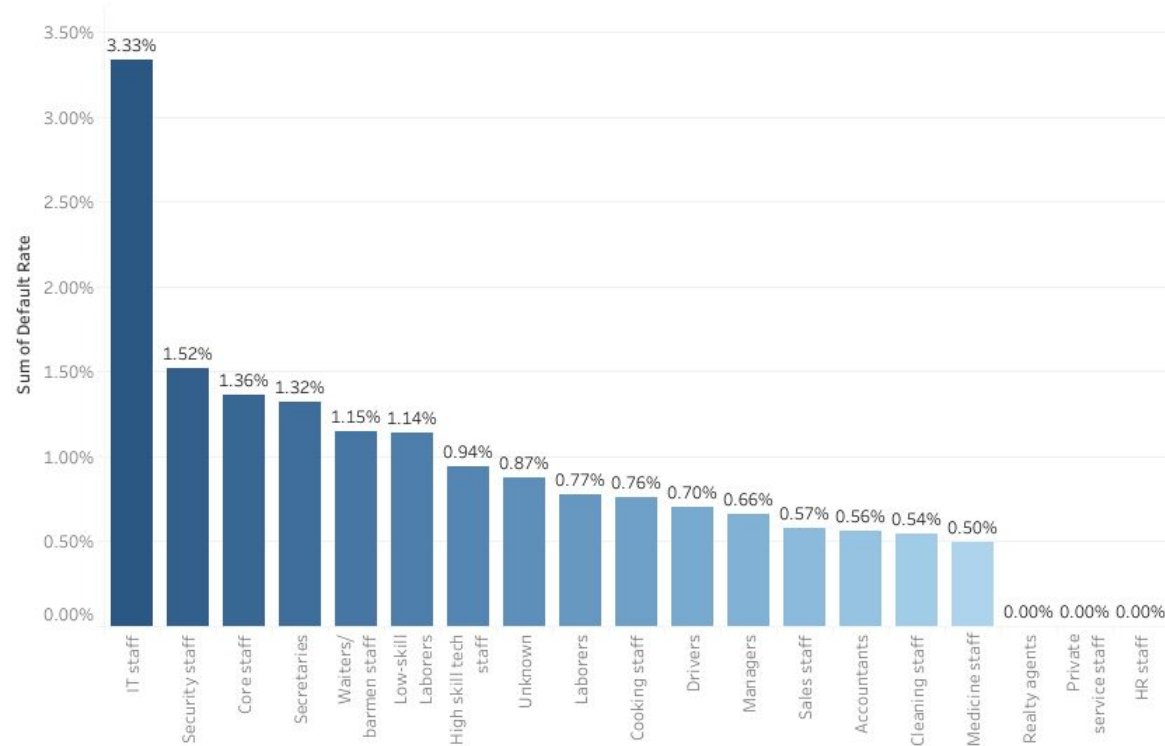


- Default risk **decreases** systematically as **education level increases**.
- Applicants with **lower or incomplete education** exhibit **nearly 2–3x higher default rates** than those with **higher or academic degrees**.

Implication: underwriting can incorporate education as a supporting feature to improve risk ranking. Applicants with lower/incomplete education should trigger tighter affordability/verification checks and more conservative initial limits or risk-based pricing

“Occupation Strongly Differentiates Default Risk”

Default Rate by Occupations



Key insights

- Default risk **varies meaningfully by occupation**, creating a clear high-risk tail.
- A small set of roles shows **several-fold higher default** than the **lowest-risk occupations**.

Implication

Use occupation as a risk-segmentation signal: apply tighter limits/pricing and added verification for high-risk roles, while monitoring fairness.

Hypothesis Testing

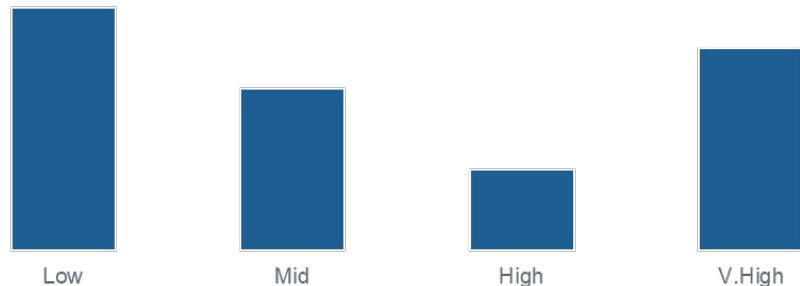
Income Is Not a Statistically Significant Driver of Default Risk

Key Takeaway

Default rates are **similar across income segments** within approved applicants.

Statistical test shows **no meaningful separation** by income alone

Default rate by income segment (illustrative)



Mann-Whitney U test

p = 0.658

Not significant

Income bands **should not drive approvals.**

Use income mainly for **limits/pricing, and prioritize stronger risk signals** (employment stability, affordability).

Employment Status Is a Statistically Relevant Driver of Default Risk

Key Takeaway

Unemployment is **associated with higher default risk**, even within approved applicants.
Employment status **meaningfully separates risk beyond income/age**.



Chi-Square Test

$p = 0.05$

Significant

Prioritize employment stability in underwriting: **add stricter checks and more conservative initial limits/pricing** for unemployed applicants.

Real Estate Ownership Lowers Default Probability

Key Takeaway

There is a **statistically significant association** between real estate ownership and lower default risk ($p = 0.0056$).



Chi-Square Test

$p = 0.0056$

Significant

Incorporate property ownership as a **weighted factor** in the credit approval model to **reduce default risk exposure**.

Household size is not a statistically significant driver of default risk among approved applicants

Key Takeaway

Despite visual differences, **household size does not significantly explain default behavior ($p = 0.688$).**



Mann-Whitney U Test

$p = 0.687$

Not Significant

Household size should not be used as a primary screening rule, as it does not significantly improve default risk differentiation.

Matrix Table

(Income x Employment x Family Size)

Risk Concentration Segments

⚠️ High Risk Profiles

Income Low	Family Large	Status Employed	3.48% Default
Income High	Family Small	Status Unemployed	1.54% Default
Income Low	Family Small	Status Unemployed	1.57% Default
Income High	Family Small	Status Employed	0.99% Default

🛡️ Low Risk Profiles

Income Very High	Family Small	Status Employed	0.09% Default
Income Mid	Family Small	Status Employed	0.69% Default
Income High	Family Medium	Status Employed	0.86% Default

KEY DRIVERS



Unemployment = Highest Risk Lift



Low Income + Large Family = Stress Spike



Employed + Small Family = Lowest Risk

Summary Insights and Recommendations



Income Level



Low Income High Income

"Flat" risk curve. High income does not guarantee repayment behavior.

RECOMMENDATION

Stop tightening income thresholds. Focus on stability instead.



Employment Status



Major Risk Gap: Unemployed applicants are 33% more likely to default.

ACTION REQUIRED

Implement immediate stricter cut-offs for unemployed applicants to filter unstable profiles.



Household Size



Larger families face higher repayment stress due to living costs.

POLICY UPDATE

Add "Family Size" as a weighted factor in the approval scorecard.

Summary Insights and Recommendations



Education Level



Lower education correlates with lower financial literacy and income stability.

🔍 ACTION REQUIRED

Apply deeper credit analysis for low-education applicants to ensure solvency.



Risk Combinations

Single-variable rules miss high-risk pockets. Risk is multiplicative.



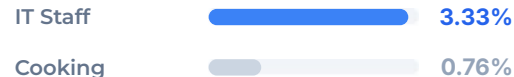
Highest Risk Segment: One-dimensional rules are currently approving these high-risk profiles.

🔄 STRATEGIC PIVOT

Implement Segment-Based Rules (Income × Employment × Family) to cut defaults by ≥10%.



Occupation Type



Volatile income behavior differs vastly by job type. Occupation is a key risk multiplier.

⚠️ ACTION REQUIRED

Use occupation as a precise risk multiplier/screener without reducing total approval volume.

Thank You

Does anyone have any questions?

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Appendix

Final Dashboard



Credit Risk Segmentation Dashboard

Filter



Total Applicants

36,457

Overall Default Rate

0.83%

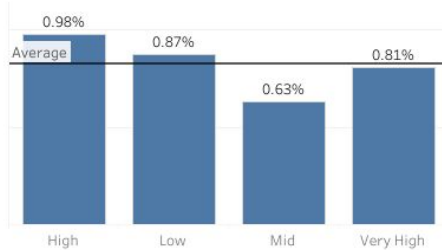
Total Bad Customers

302

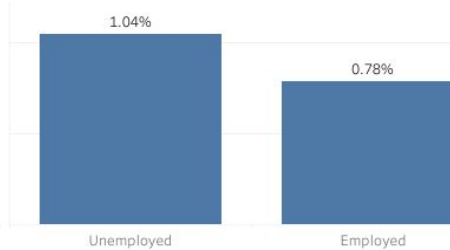
Total Good Customers

36,155

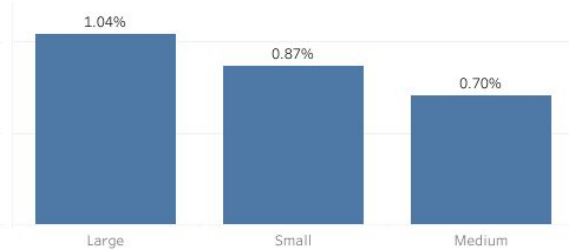
Default Rate by Income Segments



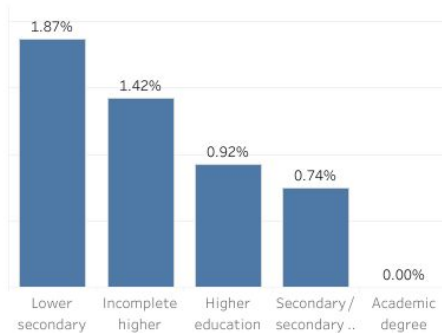
Default Rate by Employment Status



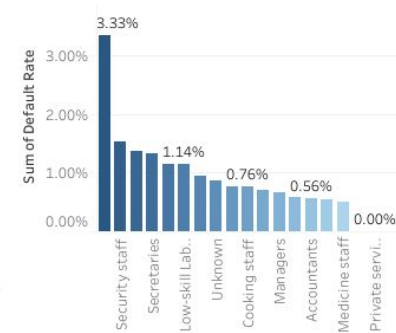
Default Rate by Family Segments



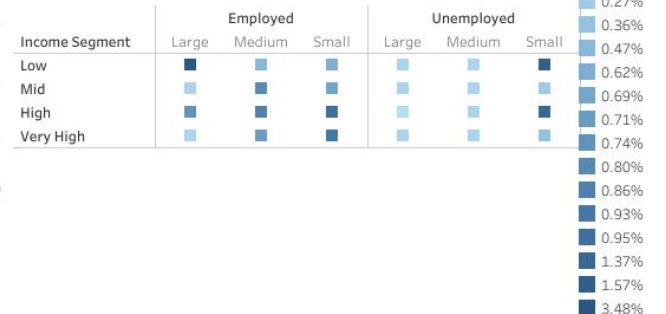
Default Rate by Education Levels



Default Rate by Occupations



Risk Matrix Heatmap (Income x Employment x Family)



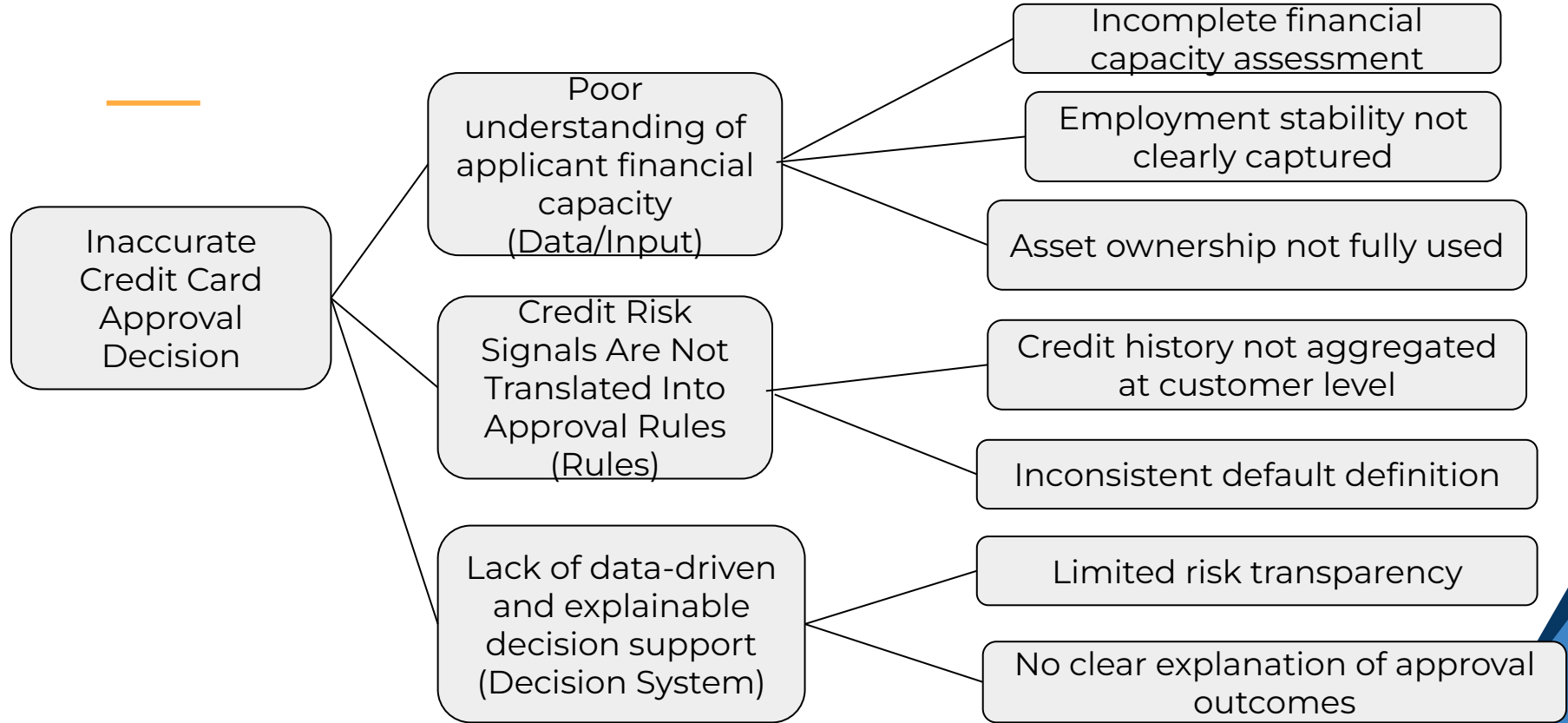
Linked Worksheet

- **Tableau Public Workbook :**
<https://drive.google.com/file/d/1N316N1GUQyW6XHbhE9gCmPxxAgkjdnsT/view?usp=sharing>
- **Google Colab Workbook :**
https://drive.google.com/file/d/1Zd_wZ4MwsK4ElQZgL96xLCY22z5sl42e/view?usp=sharing
- **Google Slides Deck :**
<https://docs.google.com/presentation/d/1NYBwKSexfjvIPQt35XFA-miAmeT7WMmlmWFWIRiSBA4/edit?usp=sharing>
- **Final Dashboard :**
https://public.tableau.com/views/TableauDEEPPRiskSegmentationWorkbook-AlbertusArielR/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

DARCI Framework

Decider (D)	Chief Risk Officer (CRO)
Accountable (A)	Credit Risk Manager
Responsible (R)	Credit Risk Analyst
Consulted (C)	Underwriting Team, Compliance, Data Engineering
Informed (I)	Business Leadership, Product, Marketing

Root Cause: Issue Tree Method



Handling Missing Values

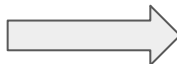
- Missing values in the **employment duration** variable were intentionally retained **to represent applicants who are currently unemployed**. In such cases, employment duration is not applicable rather than missing due to data quality issues. **A separate unemployment indicator was introduced to explicitly capture this condition.**
- There is no missing value in 'credit_record' table

EMPLOYED_YEARS	75329
CODE_GENDER	0
ID	0
FLAG_OWN_REALTY	0
CNT_CHILDREN	0
AMT_INCOME_TOTAL	0
FLAG_OWN_CAR	0
NAME_INCOME_TYPE	0
NAME_EDUCATION_TYPE	0
NAME_HOUSING_TYPE	0
NAME_FAMILY_STATUS	0
DAYS_EMPLOYED	0
FLAG_MOBIL	0
FLAG_WORK_PHONE	0
DAYS_BIRTH	0
FLAG_PHONE	0
FLAG_EMAIL	0
CNT_FAM_MEMBERS	0
OCCUPATION_TYPE	0
AGE_YEARS	0
IS_UNEMPLOYED	0
OWN_CAR	0
OWN_REALTY	0

dtype: int84

Correcting Data Types for Each Table

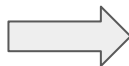
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 438557 entries, 0 to 438556
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   ID                     438557 non-null  int64
1   CODE_GENDER            438557 non-null  object
2   FLAG_OWN_CAR           438557 non-null  object
3   FLAG_OWN_REALTY        438557 non-null  object
4   CNT_CHILDREN           438557 non-null  int64
5   AMT_INCOME_TOTAL       438557 non-null  float64
6   NAME_INCOME_TYPE        438557 non-null  object
7   NAME_EDUCATION_TYPE     438557 non-null  object
8   NAME_FAMILY_STATUS      438557 non-null  object
9   NAME_HOUSING_TYPE       438557 non-null  object
10  DAYS_BIRTH              438557 non-null  int64
11  DAYS_EMPLOYED           438557 non-null  int64
12  FLAG_MOBIL              438557 non-null  int64
13  FLAG_WORK_PHONE         438557 non-null  int64
14  FLAG_PHONE              438557 non-null  int64
15  FLAG_EMAIL              438557 non-null  int64
16  OCCUPATION_TYPE         304354 non-null  object
17  CNT_FAM_MEMBERS         438557 non-null  float64
dtypes: float64(2), int64(8), object(8)
memory usage: 60.2+ MB
```



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 438557 entries, 0 to 438556
Data columns (total 23 columns):
#   Column                Non-Null Count  Dtype
---  -
0   ID                     438557 non-null  object
1   CODE_GENDER            438557 non-null  object
2   FLAG_OWN_CAR           438557 non-null  object
3   FLAG_OWN_REALTY        438557 non-null  object
4   CNT_CHILDREN           438557 non-null  int64
5   AMT_INCOME_TOTAL       438557 non-null  float64
6   NAME_INCOME_TYPE        438557 non-null  object
7   NAME_EDUCATION_TYPE     438557 non-null  object
8   NAME_FAMILY_STATUS      438557 non-null  object
9   NAME_HOUSING_TYPE       438557 non-null  object
10  DAYS_BIRTH              438557 non-null  int64
11  DAYS_EMPLOYED           438557 non-null  int64
12  FLAG_MOBIL              438557 non-null  int64
13  FLAG_WORK_PHONE         438557 non-null  int64
14  FLAG_PHONE              438557 non-null  int64
15  FLAG_EMAIL              438557 non-null  int64
16  OCCUPATION_TYPE         438557 non-null  object
17  CNT_FAM_MEMBERS         438557 non-null  int64
18  AGE_YEARS               438557 non-null  int64
19  IS_UNEMPLOYED           438557 non-null  int64
20  EMPLOYED_YEARS          363228 non-null  float64
21  OWN_CAR                 438557 non-null  int64
22  OWN_REALTY              438557 non-null  int64
dtypes: Int64(2), float64(2), int64(10), object(9)
memory usage: 77.8+ MB
```

Correcting Data Types for Each Table

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 1048575 entries, 0 to 1048574  
Data columns (total 3 columns):  
#   Column          Non-Null Count  Dtype  
---  ---  
0   ID               1048575 non-null  int64  
1   MONTHS_BALANCE  1048575 non-null  int64  
2   STATUS           1048575 non-null  object  
dtypes: int64(2), object(1)  
memory usage: 24.0+ MB
```



```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 1048575 entries, 0 to 1048574  
Data columns (total 3 columns):  
#   Column          Non-Null Count  Dtype  
---  ---  
0   ID               1048575 non-null  object  
1   MONTHS_BALANCE  1048575 non-null  int64  
2   STATUS           1048575 non-null  object  
dtypes: int64(1), object(2)  
memory usage: 24.0+ MB
```

Making Default Category per Customer from Credit Record Table

```
df_credit_record_cl['IS_DEFAULT'] = (  
    df_credit_record_cl['STATUS'].isin(['3', '4', '5'])  
).astype(int)  
  
# The worst observed credit status across all months is used to assign a single default label to each customer.  
default_per_id = (  
    df_credit_record_cl  
    .groupby('ID', as_index=False)['IS_DEFAULT']  
    .max()  
)  
  
# Default Distribution  
default_per_id['IS_DEFAULT'].value_counts(normalize=True)  
|
```

proportion	
IS_DEFAULT	
0	0.992802
1	0.007198

dtype: float64

ID DEFAULT_FLAG		
0	5001711	0
1	5001712	0
2	5001713	0
3	5001714	0
4	5001715	0
5	5001717	0
6	5001718	0
7	5001719	0
8	5001720	0
9	5001723	0
10	5001724	0

Make a join table consist of Default Label per Customer and Applicant Table

	ID	CODE_GENDER	FLAG_OWN_CAR	FLAG_OWN_REALTY	CNT_CHILDREN	AMT_INCOME_TOTAL	NAME_INCOME_TYPE	NAME_EDUCATION_TYPE	NAME_FAMILY_STATUS	NAME_HOUSING_TYPE	...	FLAG_PHONE	FLAG_EMAIL
0	5008804	M	Y	Y	0	427500.0	Working	Higher education	Civil marriage	Rented apartment	...	0	1
1	5008805	M	Y	Y	0	427500.0	Working	Higher education	Civil marriage	Rented apartment	...	0	1
2	5008806	M	Y	Y	0	112500.0	Working	Secondary / secondary special	Married	House / apartment	...	0	1
3	5008808	F	N	Y	0	270000.0	Commercial associate	Secondary / secondary special	Single / not married	House / apartment	...	1	1
4	5008809	F	N	Y	0	270000.0	Commercial associate	Secondary / secondary special	Single / not married	House / apartment	...	1	1
5	5008810	F	N	Y	0	270000.0	Commercial associate	Secondary / secondary special	Single / not married	House / apartment	...	1	1

EDUCATION_TYPE	NAME_FAMILY_STATUS	NAME_HOUSING_TYPE	...	FLAG_PHONE	FLAG_EMAIL	OCCUPATION_TYPE	CNT_FAM_MEMBERS	AGE_YEARS	IS_UNEMPLOYED	EMPLOYED_YEARS	OWN_CAR	OWN_REALTY	DEFAULT_FLAG
Higher education	Civil marriage	Rented apartment	...	0	0	Unknown	2	33	0	12.4	1	1	0
Higher education	Civil marriage	Rented apartment	...	0	0	Unknown	2	33	0	12.4	1	1	0
Secondary / secondary special	Married	House / apartment	...	0	0	Security staff	2	59	0	3.1	1	1	0
Secondary / secondary special	Single / not married	House / apartment	...	1	1	Sales staff	1	52	0	8.4	0	1	0
Secondary / secondary special	Single / not married	House / apartment	...	1	1	Sales staff	1	52	0	8.4	0	1	0
Secondary / secondary special	Single / not married	House / apartment	...	1	1	Sales staff	1	52	0	8.4	0	1	0
Secondary / secondary special	Single / not married	House / apartment	...	1	1	Sales staff	1	52	0	8.4	0	1	0